NATIONAL UNIVERSITY OF LESOTHO

BSc, B. Eng. Examinations

CS3541 Computer Communication and Networks 1

January 2024 **Marks: 100** **Time: 3 Hours**

Instructions

1. This is a *closed-book* examination.
2. **Section A is compulsory.**
3. Answer *any other 4* questions in Section B.
4. Each question carries 20 marks.
5. Each question **MUST** begin on a new page.
6. This question paper consists of 8 printed pages including this cover page.

SECTION A COMPULSORY [20 MARKS]

Question 1

1.1 How does a networked server manage requests from multiple clients for different services?
(a) The server sends all requests through a default gateway.
(b) Each request has a combination of source and destination port numbers, coming from a unique IP address.
(c) Each request is tracked through the physical address of the client.
(d) The server uses IP addresses to identify different services.

1.2 Given a /32 Global Routing Prefix and a /64 prefix, how many bits would be allocated for the Subnet ID?
(a) 8
(b) 16
(c) 32
(d) 48
(e) 64

1.3 When retrieving email messages, which protocol allows for easy, centralized storage and backup of emails that would be desirable for a small- to medium-sized business?
(a) IMAP
(b) HTTPS
(c) POP3
(d) TCP

1.4 A user calls to report that a PC cannot access the internet. The network technician asks the user to issue the command **ping 127.0.0.1** in a command prompt window. The user reports that the result is four positive replies. What conclusion can be drawn based on this connectivity test?
(a) The PC can access the network. The problem exists beyond the local network.
(b) The TCP/IP implementation is functional.
(c) The PC can access the Internet. However, the web browser may not work.
(d) The IP address obtained from the DHCP server is correct.

1.5 What TCP mechanism is used to enhance performance by allowing a device to continuously send a steady stream of segments as long as the device is also receiving necessary acknowledgements?
(a) IP address/port pair
(b) 3-way handshake
(c) 2-way handshake
(d) Sliding window

1.6 Which statement is true about variable-length subnet masking?
(a) Bits are returned, rather than borrowed, to create additional subnets.
(b) Subnets may only be subnetted one additional time.
(c) The size of each subnet may be different, depending on requirements.
(d) Each subnet is the same size.

1.7 RA messages are sent to all IPv6 routers by hosts requesting addressing information.
(a) False
(b) True

1.8 What kind of threat is described when a threat actor overloads a network to deny other users network access?
(a) Data manipulation
(b) Identity theft
(c) Information theft
(d) Denial of service

1.9 The **Date:** header in the HTTP response message indicates when the object in the response was last modified.
(a) True
(b) False

1.10 Network congestion has resulted in the source learning of the loss of TCP segments that were sent to the destination. What is one way that the TCP protocol addresses this?
(a) The source decreases the amount of data it transmits before it receives an acknowledgment from the destination.
(b) The destination decreases the window size.
(c) The source decreases the window size to decrease the rate of transmission from the destination.
(d) The destination sends fewer acknowledgments in order to conserve bandwidth.

1.11 Source host A sends a large file to the destination host B over a TCP connection. If the sequence number of this connection is *k*, then the sequence number for the subsequent segment will be *k + 1*.
(a) True
(b) False

1.12 What process is used to place one message inside another message for transfer from the source to the destination?
(a) Flow control
(b) Access control
(c) Layering
(d) Encapsulation
(e) Decoding

1.13 Which procedure is used to reduce the effect of crosstalk in copper cables?
(a) Wrapping the bundle of wires with metallic shielding.
(b) Twisting opposing circuit wire pairs together.
(c) Avoiding sharp bends during installation.
(d) Designing a cable infrastructure to avoid crosstalk interference.
(e) Requiring proper grounding connections.

1.14 A network administrator is designing a new network infrastructure that includes both wired and wireless connectivity. Under which situation would a wired connection be recommended?
(a) When the end-user device needs mobility when connected to the network.

(b) When the end-user device requires a dedicated connection because of performance requirements.
(c) When the end-user device only has wireless Ethernet NIC.
(d) When the organization uses BYOD policy.

1.15 What method is used to manage contention-based access on a wireless network?
(a) Token passing
(b) CSMA/CD
(c) CSMA/CA
(d) Priority ordering

1.16 Which statement describes the full-duplex mode of data transmission?
(a) Data that is transmitted over the network flows in one direction to many different destinations simultaneously.
(b) Data that is transmitted over the network flows in one direction at a time.
(c) Data that is transmitted over the network flows in both directions at the same time.
(d) Data that is transmitted over the network can only flow in one direction.

1.17 Although CSMA/CD is still a feature of Ethernet, why is it no longer necessary?
(a) It has been replaced by CSMA/CA.
(b) The use of full-duplex capable Layer 2 switches.
(c) The use of Gigabit Ethernet speeds.
(d) The virtually unlimited availability of IPv6 addresses.
(e) The development of half-duplex switch operation.

1.18 Which field in an IPv6 packet is used by the router to determine if a packet has expired and should be dropped?
(a) No Route to Destination
(b) Packet Length
(c) Time-to-Live
(d) Hop Limit
(e) Address Unreachable

1.19 When a router receives a packet, what information must be examined in order for the packet to be forwarded to a remote destination?
(a) Source MAC address
(b) Destination MAC address
(c) Source IP address
(d) Destination IP address
(e) Both Source and Destination IP addresses

1.20 What is one function of the ARP protocol?
(a) Resolving an IPv4 address to a MAC address.
(b) Resolving both IPv4 and IPv6 addresses to a MAC address.
(c) Maintaining a table of domain names with their resolved IP addresses.
(d) Mapping a domain name to its IP address.
(e) Automatic assignment of IP addresses.

[20 marks]

SECTION B ANSWERS ANY 4 QUESTIONS [80 MARKS]

Question 2

(a) Suppose the source host sends two TCP segments back-to-back to the destination host over a TCP connection. The first segment has SEQ number 87; the second segment has SEQ number 104. **(2)**

(i) How much payload is in the first segment?
(ii) Suppose the first segment is lost but the second segment arrives at the destination. What will be the ACK number that the destination host sends to the source host? Explain. **(2)**

(b) Discuss any two responsibilities of the transport layer in the TCP/IP model. **(2)**

(c) Suppose Lerato uses her web-based Gmail account to send a message to Thabo, who accesses his mail from a mail server using POP3. Explain how the email message moves from Lerato's host to Thabo's host by listing the series of application-layer protocols used to move the message between these two hosts. **(3)**

(d) Suppose you wanted to do a transaction from a remote host fast. Which transport layer protocol would you use and why? **(2)**

(e) You are given a host IPv4 address, the original subnet mask, and the new subnet mask.

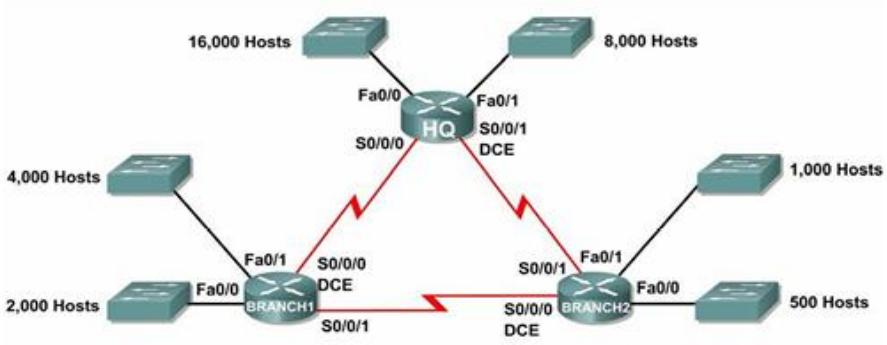
Host IP Address:	128.107.0.55
Original Subnet Mask	255.255.0.0
New Subnet Mask:	255.255.255.0

(i) Write the original subnet mask using slash notation. **(1)**
(ii) Calculate the number of subnet bits and the number of host bits per subnet **(1)**.
(iii) Calculate the number of subnets, and number of usable hosts per subnet **(1)**.
(iv) Subnet this IP address and list the IP addresses for the first eight subnets using prefix notation. **(4)**
(v) For the eighth subnet, write down the IPv4 address for the first and last host addresses using prefix notation **(2).**

[20 marks]

Question 3

(a) Apply variable-length subnetting to segment the **172.16.128.0/17** network to successfully fulfill the IP addressing requirements of the given topology. Show your work and any assumptions you make. **(14)**



(b) Using the common IP addressing scheme, assign IPv4 addresses to all router interfaces and the switch to router interfaces. Present your answer in a table similar to the one below: **(6)**

Device	**Interface**	**IP Address**	**Subnet Mask**
HQ	S0/0/0		
	S0/0/1 (DCE)		
	Fa0/0		
	Fa0/1		
BRANCH-1	S0/0/0 (DCE)		
	S0/0/1		
	Fa0/0		
	Fa0/1		
BRANCH-2	S0/0/0 (DCE)		
	S0/0/1		
	Fa0/0		
	Fa0/1		

[20 marks]

Question 4

(a) Which transport layer protocol would you prefer for VoIP and IP video conferencing applications? Why? **(3)**
(b) State two features of ICMPv6 that are not present in ICMPv4. **(2)**
(c) Explain what ICMP is and give examples of its key areas of functionality. **(3)**
(d) Discuss the issues and solutions related to the combining of IPv6 and IPv4 addresses for digital network communications. **(9)**
(e) Briefly explain how the DNS is implemented globally and how DNS queries are resolved. **(3)**

[20 marks]

Question 5

(a) The network layer in the source host accepts a segment of 1,200 bytes and a destination host address from the transport layer. Suppose many network application processes run at the destination host at the same time.

(i) Design a transport-layer protocol that will get application data to the desired process at the destination, assuming the operating system in the destination host has assigned a 4-byte port number to each running application process. **(5)**

(ii) Modify the protocol in (i) so that it provides a return address to the destination process. **(5)**

(iii) Describe the EUI-64 process and explain what it is used for. **(6)**

(iv) State and explain one dynamic addressing method for the configuration of IPv6 GUAs. **(4)**

[20 marks]

Question 6

(a) The National University of Lesotho intends to connect one of its campuses to the main campus network. The service provider has the following options: Dedicated Leased Line, ADSL, and Satellite.

(i) Choose the appropriate option for the NUL and explain your choice. **(2)**

(ii) For technologies that were not chosen, explain why they were deemed not so appropriate. **(2)**

(b) A company's network can be breached, giving threat actors access to the confidential information of the organization. For this reason, network security is always going to be a top priority for administrators.

(i) Differentiate between a Trojan horse and a virus. (**2)**

(ii) It is important to consider both external and internal attacks. Differentiate between the two categories of attacks. **(2)**

(c) Cisco devices offer multiple access methods. Give commands that are used to secure each of the following access methods or modes assuming the device is already in the Global Configuration Mode:

(i) Console line **(2)**

(ii) Telnet **(2)**

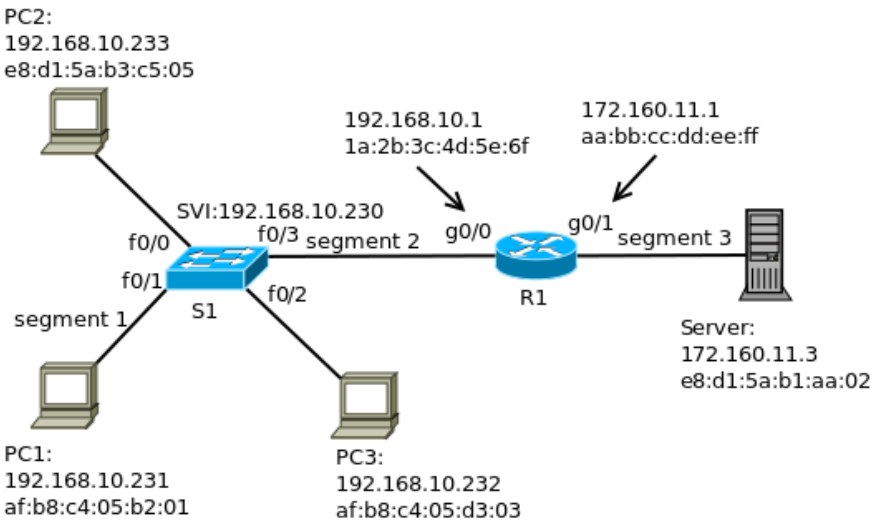
(iii) Privilege EXEC mode **(1)**

(d) In computer networks, frames can be delivered in different ways. Differentiate between unicast and broadcast delivery options. Also, give an example frame destination address for each option. **(4)**

(e) A web client is sending a request for a webpage to a web server. From the perspective of the client, what protocols (in the correct order) of the TCP/IP protocol stack (Application to Datalink) are used to prepare the request for transmission? **(3)**

[20 marks]

Question 7



Consider the topology shown in the figure above. Assume frames are sent from PC1 to other hosts in the network;

(a) Indicate the values of the source and destination addresses (both MAC and IP) for segment 2 and segment 3. **(4)**

(b) Show the CAM table of S1, assuming that all hosts have communicated. **(4)**

(c) Briefly explain the role of ARP for communication between PC1 and PC2, and between PC1 and the Server. **(4)**

(d) Suppose S1 is configured to use a Store-and-Forward switching method. Give one advantage and one drawback of this switching method. **(2)**

(e) Briefly explain how S1 switches packets. **(4)**

(f) Give the type of cables that have been used to connect S1 to R1 and R1 to the server. Explain. **(2)**

[20 marks]